MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 12 (8-12 Nov 2026) - Period 1	Ch 8 - 8.8 Totalling and 8.9 Counting

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 82% wrote correct REPEAT validation. Today: totalling and counting inside loops, linking to Chapter 7 standard algorithms.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use totalling (Total <- Total + Value) and counting (Count <- Count + 1) inside loops, and calculate averages. Syllabus 8.1.4(d); AO2, AO3.

Success criteria — *what will successful learning look / sound like?*

I initialise totals and counters to 0 before the loop; I update them correctly inside the loop; I output an accurate total, count and average.

Key knowledge / skill / understanding:

Totalling adds values; counting adds 1 when a condition is met; initialise before the loop, output after it.

Likely misconception / access or challenge need:

Initialise inside the loop; average = total / loop counter even when skipping values.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Problem solving **Time:** 8 min

What students will do:

Count and total by hand: a list of 10 temperatures - how many above 30 and what is the total and average? Write the algorithm steps.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Data slide.

Support (low performers / SEN):

Calculator allowed for warm-up only.

Stretch (fast learners / gifted):

Explain why average must be REAL.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Project-based task **Time:** 22 min

What students will do:

Class charity sale program: input item prices until 0, output total, number of items, items over 20 SAR and average price; pseudocode first, then Python.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.8-8.9; Python.

Support (low performers / SEN):

Skeleton with loop and comments.

Stretch (fast learners / gifted):

Track the most expensive item (link to finding maximum).

ACTIVITY 3 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 7 min

What students will do:

Trace table of a counting/totalling algorithm with 5 inputs; compare with partner and fix differences.

Evidence tool: Exit ticket

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Trace table sheet.

Support (low performers / SEN):

Trace table with columns labelled.

Stretch (fast learners / gifted):

Add a column for a running average.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: the school charity sale - how much did we raise and how many items sold? Model Total <- 0, Count <- 0 before a loop, updates inside, OUTPUT after. Show Python equivalent and integer vs real average.

What is new or innovative in this lesson? *(ETEC looks for this)*

Real school charity context; the same program is designed, coded and traced, mirroring Paper 2 scenario-question skills.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Resetting total inside the loop; counting the sentinel 0; dividing by zero when no items; integer division for average. Likely: 'my total is always the last value'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: write the two lines that total and count; identify where initialisation belongs. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.8-8.9 questions; program: input 10 marks, output total, count of passes (>= 50) and average.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Charity context resonated; students added cultural items to the sale list. 80% got total, count and average right.

What will I change next time?

Show the sentinel-value bug live; several students counted the 0.

Students needing follow-up:

10BR2: 3 students reset totals inside loops; quick re-teach with a trace table.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☐1 ☐2 ☑3 ☑C ☐H	☑1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H
Create	**Discussion**	**Pair / group**	**Learner choice**	**Self-assessment**	**Peer assessment**	**SEN / low**	**Gifted**
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 12 (8-12 Nov 2026) - Period 2	Ch 8 - 8.10 String handling

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 12 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Charity programs: 80% correct; sentinel counting errors fixed by most. Today: string handling LENGTH, SUBSTRING, UCASE, LCASE.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use LENGTH, SUBSTRING, UCASE and LCASE in pseudocode and Python (len, slicing, upper, lower), noting position 1 vs 0. Syllabus 8.1.4(e); AO1, AO2.

Success criteria — *what will successful learning look / sound like?*

I can predict the result of LENGTH, SUBSTRING, UCASE, LCASE; I can convert SUBSTRING (s, start, length) into Python slicing correctly; my username generator works.

Key knowledge / skill / understanding:

Strings are sequences of characters; pseudocode SUBSTRING starts at 1 in the syllabus examples, Python indexes from 0.

Likely misconception / access or challenge need:

SUBSTRING(s,1,5) = s[1:5] in Python; LENGTH counts only letters, not spaces.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Inquiry / investigation **Time:** 10 min	**Strategy:** Problem solving **Time:** 20 min	**Strategy:** Gamification **Time:** 7 min
What students will do: Investigate in the Python shell: len(), slicing, upper(), lower() on students' own names (with consent); record results and derive rules.	**What students will do:** Username generator: first 3 letters of surname + first letter of first name + birth year last 2 digits, all lowercase; write pseudocode with SUBSTRING/LCASE, then Python.	**What students will do:** Kahoot 'String surgeon': predict outputs of 8 string expressions in pseudocode and Python.
Evidence tool: Portfolio / notebook work Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group	**Evidence tool:** Practical / performance task Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Digital quiz (Forms / Kahoot / Quizizz) Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group
Resources: Investigation sheet; Python shell.	**Resources:** Coursebook (CUP 9781108915144) 8.10; syllabus string functions p.42.	**Resources:** Kahoot.
Support (low performers / SEN): Sheet with examples completed.	**Support (low performers / SEN):** Worked SUBSTRING-to-slice conversion card.	**Support (low performers / SEN):** Questions with images.
Stretch (fast learners / gifted): Find how to get the last 3 characters in 2 ways.	**Stretch (fast learners / gifted):** Handle names shorter than 3 letters; add validation.	**Stretch (fast learners / gifted):** Bonus: write the expression for a given output.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: school email usernames - how are they made? Model LENGTH("Happy Days") = 10, SUBSTRING("Happy Days", 1, 5) = "Happy", UCASE/LCASE (syllabus examples), then Python len, s[0:5], upper(), lower().

What is new or innovative in this lesson? *(ETEC looks for this)*

Names from different languages (Arabic transliteration, hyphenated names) become test data, showing why robust string handling matters.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Off-by-one in slicing; forgetting strings are immutable; names with Arabic script or spaces/hyphens. Likely: 'SUBSTRING(1,5) = [1:5]'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: convert SUBSTRING(Word, 2, 3) into Python; predict UCASE("riyadh"). Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.10 questions; program: count vowels in a sentence (uses loop + string handling).

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Using students' own names (with consent) made strings personal; transliterated names created great test cases. 80% converted SUBSTRING correctly.

What will I change next time?

Teach 1-based vs 0-based indexing with a visual ruler before coding.

Students needing follow-up:

10BR1: 3 students confused about slicing end index; extra practice sheet.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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